



Instructor: Date:

Learner: Regional Branch:

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Let's Start!

Events and Delegates to Connect Commands



Learning Outcome:

Describe a general-purpose computer programming language in spoken and written situations.

Ask your partner:

- What is the best software to create web apps?
- What do you think is the process behind a click?

Can Do Statements:

- I can provide information about events and event-driven applications in the .NET framework.
- I can talk about the role of delegates in the .NET framework.
- I can talk about characteristics for delegates in the .NET framework.





Let's Explore!

Raul is asking for some advice to his classmate Ethan. What are they talking about?



Instructions: Read the text below.

Raul: Ethan, can I ask you something?

Ethan: Sure, shoot.

Raul: I have to make this app communicate to the mail server to send daily newsfeed to the subscribers as the teacher asked us to, but I don't remember how to do it.

Ethan: Well, that can be the easiest thing depending on what framework you use, what are you going to use to do it?

Raul: Well, I think I'll use Oracle, it is the one I remember the most.

Ethan: Why aren't you going to use .NET? I think it is the wisest option, as you are working with Windows.

Raul: Well, I haven't thought about it, but now that you mention it, that could be a really good idea.

Ethan: Say no more. Let's see, I think the most intelligent way to do it is by using delegates as it's an event-driven application.

Raul: Wait, what?

Ethan: Hmmm, I see, you didn't come to that class. Let's imagine that every time you click on a button, it automatically generates an event or action, that is a line of code one can handle; and an event-driven application is an app that executes or invokes code as a response to any event.

Raul: I see.

Ethan: Therefore, as in your case, where you need to concatenate loose objects, the best way to do so is by using delegates. They bind pointers that lead to specific code. The fact of clicking over a submit button after having filled a form, for example, generates an event, but that event needs a delegate that equates to a function pointer, such pointer only references to a specific function, making it type-safe and secure. Consequently, a delegate consists of a function pointer heading to a specific object, and references to one or many other methods within the object.



Let's Practice!

Are you going to use an event-driven application?

Instructions: Read the sentences Underline all the mistakes in the paragraph.

1. Aren't you going to using an event-driven application?
2. What are you go to do to connect the mail server in order to send daily newsfeed to the subscribers?
3. Is Oracle what your going to use?
4. Why are you going to works with that instead of using an event-driven application?
5. What are going to happen if you click over a submit button when using an event-driven application?



Let's Practice!

Could you define and describe delegates in the .Net Framework?

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. *The most intelligent* way to do it is by using delegates as it's an event-driven application.
2. Delegates are *common* classes used in the .Net Framework.
3. Function pointers are *recurrent* variables that work along with delegates in the .Net Framework.
4. When the binding mechanism uses delegates, it turns out to be *the most dynamic* method.
5. Delegates could be *the most multicasting* classes.
6. Delegates are *useful* as they use a bound method to communicate with other classes.

Adjectives	Superlatives



Extension Activity

Diagrams and delegates in the .Net Framework

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about the role of delegates in an event-driven application in the .Net Framework.
- Draw a diagram in which you describe the role of delegates in an event-driven application.
- Show your diagram to your partners and talk about the way delegates work.
- Check your pronunciation with your instructor.

Self-Assessment

Yes No

- I look for further information about the role of delegates in the .Net Framework.
- I draw a diagram based on the information I looked for.
- I show the diagram to my partners, talk about it and check pronunciation with my instructor.



Target Words

- **newsfeed** (noun): newest information that changes day by day in a webpage
e.g. I have to make this app communicate to the mail server to send daily newsfeed to the subscribers.
- **invoke** (verb): to put something into use, like a code into a different line of code
e.g. An event-driven application is an app that executes or invokes code as a response to any event.
- **concatenate** (verb): to link
e.g. Therefore, as in your case where you need to concatenate loose objects, the best way to do so, is by using delegates.
- **loose objects** (noun): objects that belong to a loosely coupled system, that is objects that have no knowledge of the other objects they are linked to
e.g. Therefore, as in your case where you need to concatenate loose objects, the best way to do so, is by using delegates.
- **equate** (verb): that event needs a delegate that equates to a function pointer
e.g. That event needs a delegate that equates to a function pointer.
- **function pointer** (noun): a pointer supported by object oriented technologies
e.g. A delegate consists of a function pointer heading to a specific object.



Learning Points

The Future With Going to in Questions

We use **going to** in the interrogative form to ask about future plans, actions or events.

First of all, let's have a quick review of the general structure. Remember that plans and arrangements are commonly expressed using this structure:

Subject + verb to be + going to + main verb + complement

Let's see an example of this:

*My assistant **is going to call** the software manager tomorrow.*

Now, that's the structure for affirmative statements; the negative structure is similar, with only one difference: we add the particle not after the verb to be. But right now, we will focus on the **interrogative form**, which allows us to **ask about plans and arrangements for the future**. The structure of the interrogative form that we use for this purpose is:

Verb to be + subject + going to + main verb + complement +?

Something important to notice is that this type of questions are **close-ended**; this means that the answer to them is usually **yes/no/I don't know**. Let's take a look at a couple of interactions where this structure is used:

Examples:

*A: **Are you going to reinstall** the antivirus software?*

*B: **Yes**, I am, but not today; maybe tomorrow morning.*

*A: **Is** this software update **going to support** our old drivers?*

*B: **I don't know**, we should ask the IT manager.*

As you know, there are also questions that need a full sentence as an answer; these are called wh-questions. This is the grammatical structure used in this case:

Interrogative adverb (what, where, when, etc.) + verb to be + subject + going to + main verb + complement +?

Examples:

*A: **What are you going to use** for designing the website?*

*B: **I'm going to use** an event-driven programming.*





Learning Points

Examples

- *What are you going to use to do it?*
- *Why aren't you going to use .NET?*

Superlatives

We use the **superlative** form of adjectives to express the extreme point of a quality, feature or characteristic.

At this point, you are already familiar with adjectives; they are words that express **features, qualities and characteristics** of nouns:

*The computers in this office are very **old**.*
*There is a **big** update coming next year.*

And, of course, sometimes we want to **compare** two or more people or things based on one of their common characteristics, features or qualities. For that purpose, we use comparative and superlative forms of adjectives. Today, we will focus on **superlatives**; let's take a look at the chart below to learn the superlatives of some common adjectives:

1-syllable Adjectives		
Adjective	Superlative form	Example
Fast	-est	<i>.NET is the fastest way to code.</i>
Big	The... g -est	<i>Coding is not the biggest problem you have.</i>
2-syllable adjectives		
Simple	The...-est	<i>That's the simplest program you can ever find.</i>
planned	The most...	<i>This is the most planned event on history of IT.</i>
3+ syllable adjectives		
Important	The most...	<i>.NET is one of the most important frameworks at the moment of choosing.</i>



Learning Points

Superlatives are useful when we want to express that a noun has the extreme point of a quality, compared to other nouns.

As you may have noticed, some superlatives are formed with the suffix **-est**, and others with the words **the most** before the adjective. There is a rule that you can learn to form the superlative of the vast majority of adjectives:

Adjectives with one and two syllables= The (adjective)-est

Adjectives with three or more syllables= The most (adjective)

Examples:

Jennifer is **the most intelligent** programmer I know.

This cloud service software is **the safest** in the industry right now.

Long words of the English language:

procrastination = 15 letters
defibrillation = 14 letters
incomprehensibilities = 21 letters
sesquipedalianism = 17 letters

- We can say that *incomprehensibilities* is **the longest word** of the list.
- We can also say that *defibrillation* is **the shortest word** of the list.

Examples

- I think it is the **wisest** option, as you are working with Windows.
- I think the **most intelligent** way to do it is by using delegates as it's an event-driven application.
- (...) the **best** way to do so is by using delegates.
- Well, that can be the **easiest** thing depending on what framework you use.